

Lesson 9: Multiple Logistic Regression

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Learning Objectives

1. Construct and fit a multiple logistic regression model
2. Test for significance of individual coefficients or sets of coefficients in multiple logistic regression
3. Estimate the predicted probability of our outcome using multiple logistic regression
4. Present the odds ratios for multiple variables at once
5. Interpret odds ratios for coefficients while adjusting for other variables

Late stage breast cancer example

- For breast cancer diagnosis example, recall:
 - **Outcome:** early or late stage breast cancer diagnosis (binary, categorical)
- **Primary covariate:** Race/ethnicity
 - Non-Hispanic white individuals are more likely to be diagnosed with breast cancer
 - But POC are more likely to be diagnosed at a later stage
- **Additional covariate:** Age
 - Risk factor for cancer diagnosis
 - Might be a confounder for race/ethnicity
- We want to fit a multiple logistic regression model with both risk factors included as independent variables

Learning Objectives

1. Construct and fit a multiple logistic regression model
2. Test for significance of individual coefficients or sets of coefficients in multiple logistic regression
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Introduction to Multiple Logistic Regression

- In multiple logistic regression model, we have > 1 independent variable

- Sometimes referred to as the “multivariable regression”

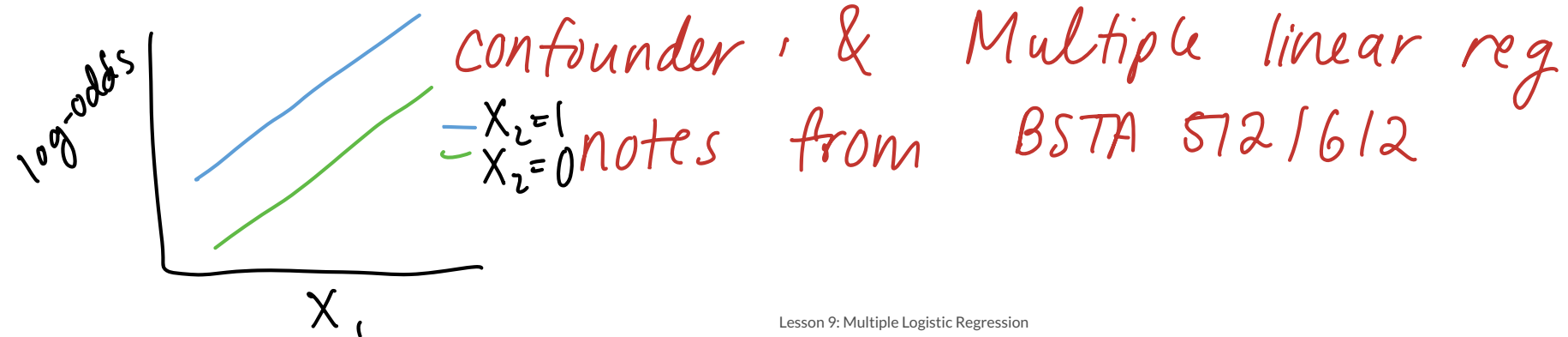
- The independent variable can be any type:

- Continuous
- Categorical (ordinal or nominal)

- We will follow similar procedures as we did for simple logistic regression

- But we need to change our interpretation of estimates because we are adjusting for other variables

or multiple =
 > 1 covariate/
 predictor/
 ind variable
multivariate =
 multiple outcomes



Multiple Logistic Regression Model

- Assume we have a collection of k independent variables, denoted by $\mathbf{X} = (X_1, X_2, \dots, X_k)$

$\uparrow \uparrow \uparrow$

given full vector
of
variables

- The conditional probability is $\underline{P(Y = 1 | \mathbf{X})} = \pi(\mathbf{X})$

Why the bold $\mathbf{X}$?

$\mathbf{X}$ represents the vector of all the X 's. This is how we represent our group of covariates in our model.

- We then model the probability with logistic regression:

$$\underline{\text{logit}(\pi(\mathbf{X}))} = \beta_0 + \beta_1 \cdot \underline{X_1} + \beta_2 \cdot \underline{X_2} + \beta_3 \cdot \underline{X_3} + \dots + \beta_k \cdot \underline{X_k}$$

$$\begin{aligned} \pi(\mathbf{X}) &= \pi(\text{race/ethnicity, age}) \\ &= P(Y=1 \mid \text{race/ethnicity, age}) \end{aligned}$$

Fitting the Multiple Logistic Regression Model

- For a multiple logistic regression model with k independent variables, the vector of coefficients can be denoted by

bolded

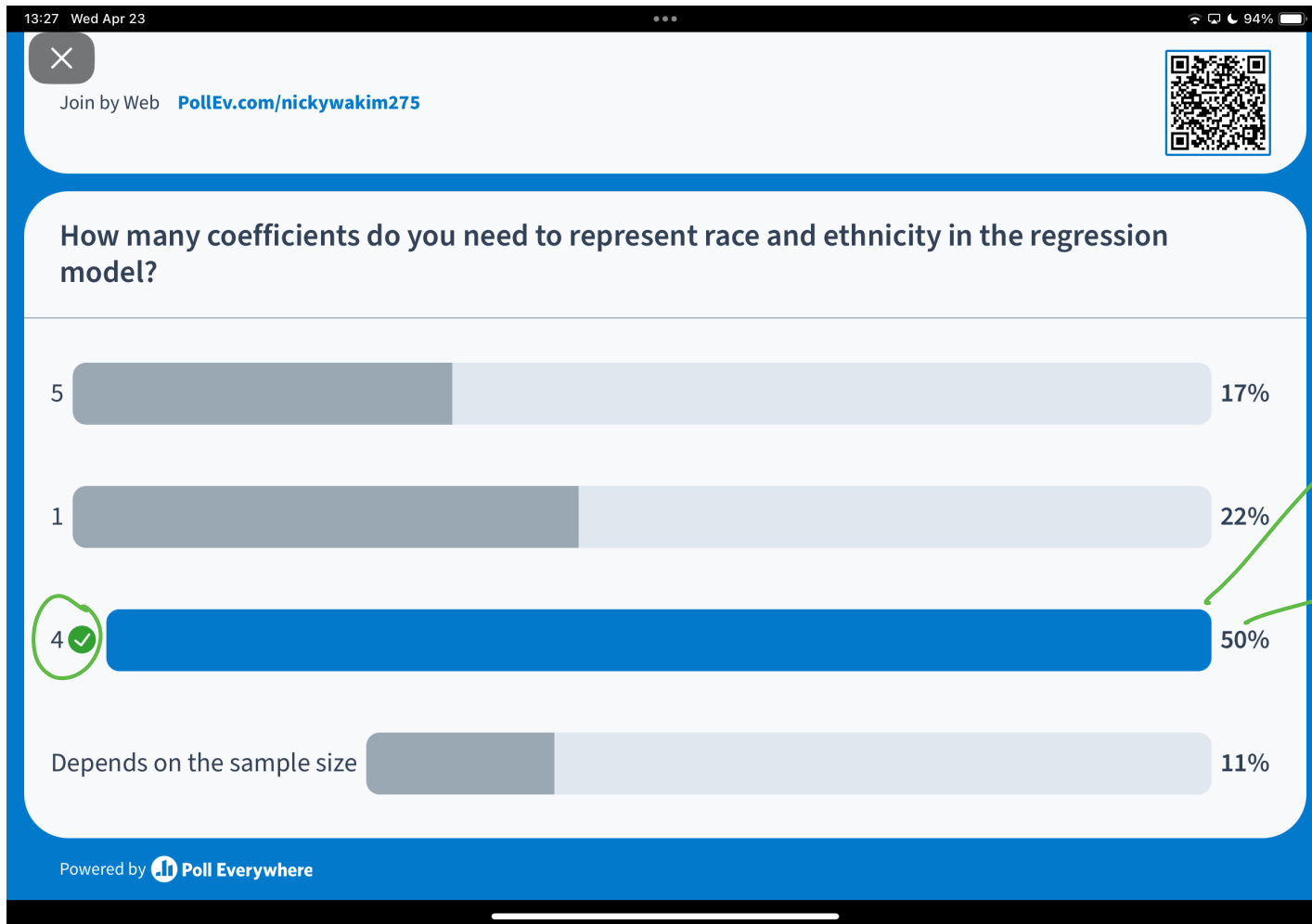
$$\underline{\beta^T} = (\beta_0, \beta_1, \beta_2, \dots, \beta_k)$$

- As with the simple logistic regression, we use **maximum likelihood method** for estimating coefficients
 - Vector of estimated coefficients:

$$\hat{\beta}^T = (\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \dots, \hat{\beta}_k)$$

- For a model with k independent variables, there is $k + 1$ coefficients to estimate
 - Unless one (or more) of those independent variables is a multi-level categorical variable, then we need more than $k + 1$ coefficients

Poll Everywhere Question 1



5 total categories
1 ref
4 coeff.
assuming intercept
& age already
in model

Breast Cancer Example: Population Model

- We can fit a logistic regression model with both race and ethnicity and age:

$$\text{logit}(\pi(\mathbf{X})) = \beta_0 + \beta_1 \cdot I(R/E = \text{"H/L"}) + \beta_2 \cdot I(R/E = \text{"NH AIAN"}) \\ + \beta_3 \cdot I(R/E = \text{"NH API"}) + \beta_4 \cdot I(R/E = \text{"NH B"}) + \beta_5 \cdot \text{Age}_c$$

- Note that race and ethnicity requires 4 coefficients to include the indicator for each category

↳ ref is NH White

- Can replace $\pi(\mathbf{X})$ with $\pi(\text{Race/ethnicity, Age})$

- 6 total coefficients (β_0 to β_5)

Fitting Multiple Logistic Regression Model: `summary()`

```
1 multi_bc = glm(Late_stage_diag ~ Race_Ethnicity + Age_c, data = bc, family = binomial)
2 summary(multi_bc)
```

Call:

```
glm(formula = Late_stage_diag ~ Race_Ethnicity + Age_c, family = binomial,
    data = bc)
```

Coefficients:

	Estimate	Std. Error	z value
(Intercept)	-1.038389	0.027292	-38.048
Race_EthnicityHispanic-Latinx	-0.015424	0.083653	-0.184
Race_EthnicityNH American Indian/Alaskan Native	-0.085704	0.484110	-0.177
Race_EthnicityNH Asian/Pacific Islander	0.133965	0.083797	1.599
Race_EthnicityNH Black	0.357692	0.071789	4.983
Age_c	0.057151	0.003209	17.811

Pr(>|z|)

(Intercept)	< 2e-16 ***
Race_EthnicityHispanic-Latinx	0.854
Race_EthnicityNH American Indian/Alaskan Native	0.859
Race_EthnicityNH Asian/Pacific Islander	0.110
Race_EthnicityNH Black	6.27e-07 ***
Age_c	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance 11861 on 9999 degrees of freedom
Residual deviance 11484 on 9994 degrees of freedom
AIC: 11496

Number of Fisher Scoring iterations: 4

model fit
statistics

★ double check what
Nicky said

Fitting Multiple Logistic Regression Model: `tidy()`

```
1 multi_bc = glm(Late_stage_diag ~ Race_Ethnicity + Age_c, data = bc, family = binomial)
2 tidy(multi_bc, conf.int=T) %>% gt() %>% tab_options(table.font.size = 38) %>%
3   fmt_number(decimals = 3)
```

term	<u>estimate</u>	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	-1.038	0.027	-38.048	0.000	-1.092	-0.985
Race_EthnicityHispanic-Latinx	-0.015	0.084	-0.184	0.854	-0.181	0.147
Race_EthnicityNH American Indian/Alaskan Native	-0.086	0.484	-0.177	0.859	-1.120	0.813
Race_EthnicityNH Asian/Pacific Islander	0.134	0.084	1.599	0.110	-0.032	0.297
Race_EthnicityNH Black	0.358	0.072	4.983	0.000	0.216	0.498
Age_c	0.057	0.003	17.811	0.000	0.051	0.063

Breast Cancer Example: Fitted Model

- We now have the **fitted logistic regression model** with both race and ethnicity and age:

fitted

$$\text{logit}(\hat{\pi}(\mathbf{X})) = \hat{\beta}_0 + \hat{\beta}_1 \cdot I(R/E = \text{"H/L"}) + \hat{\beta}_2 \cdot I(R/E = \text{"NH AIAN"}) \\ + \hat{\beta}_3 \cdot I(R/E = \text{"NH API"}) + \hat{\beta}_4 \cdot I(R/E = \text{"NH B"}) + \hat{\beta}_5 \cdot \text{Age}_c$$

$$\text{logit}(\hat{\pi}(\mathbf{X})) = -\underline{4.56} - \underline{0.02} \cdot I(R/E = \text{"H/L"}) - 0.09 \cdot I(R/E = \text{"NH AIAN"}) \\ + 0.13 \cdot I(R/E = \text{"NH API"}) + 0.36 \cdot I(R/E = \text{"NH B"}) + 0.06 \cdot \text{Age}_c$$

- 6 total coefficients ($\hat{\beta}_0$ to $\hat{\beta}_5$)

Learning Objectives

1. Construct and fit a multiple logistic regression model
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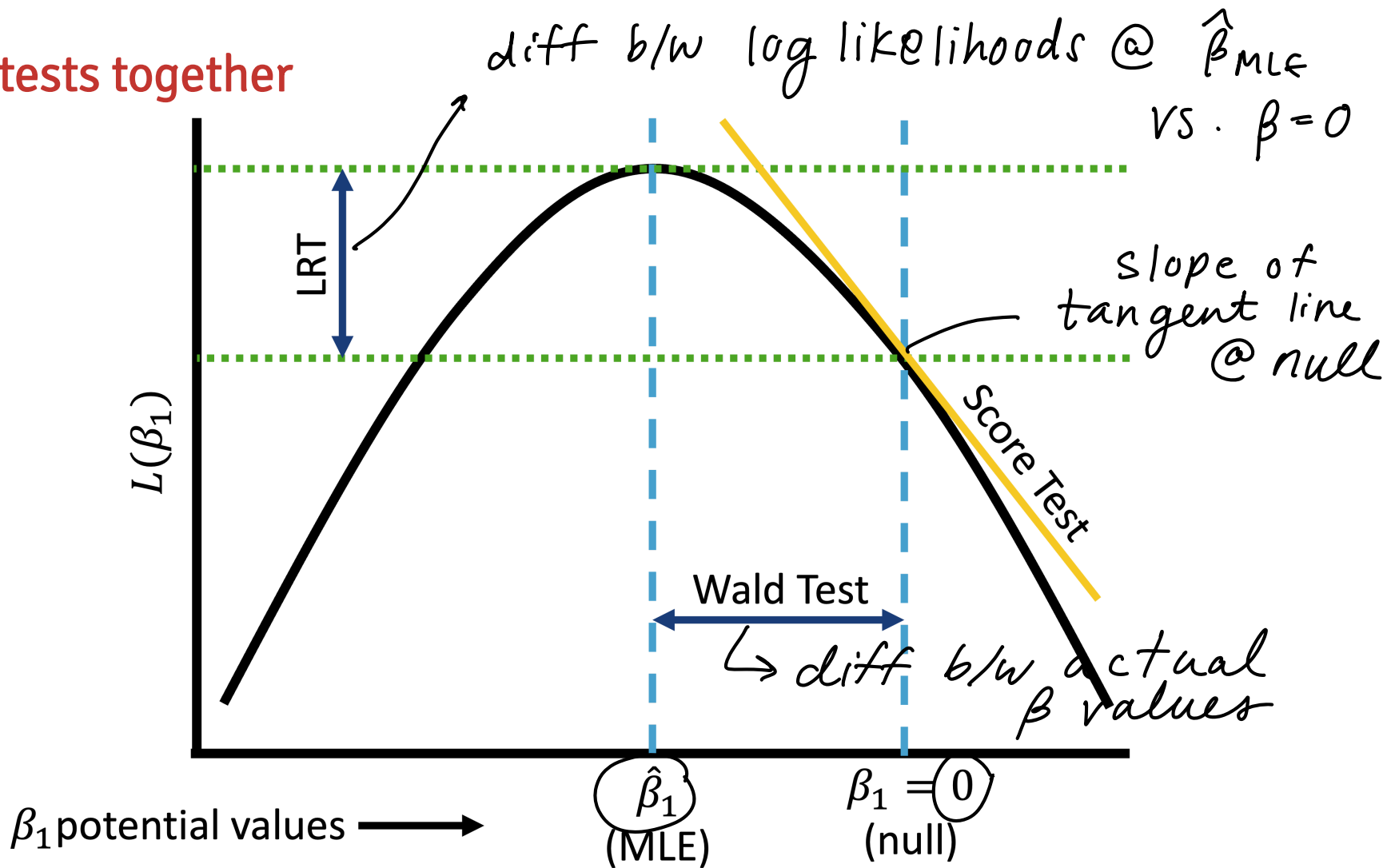
Testing Significance of the Coefficients

- Refer to Lesson 8 for more information on each test!!
- We use the same three tests that we discussed in Simple Logistic Regression to test individual coefficients
 - Wald test
 - Most common to test a single coefficient *C/*
 - ~~Score test~~
 - Likelihood ratio test (LRT)
 - Can be used to test a single coefficient or multiple coefficients
- Our class focuses on Wald and LRT only

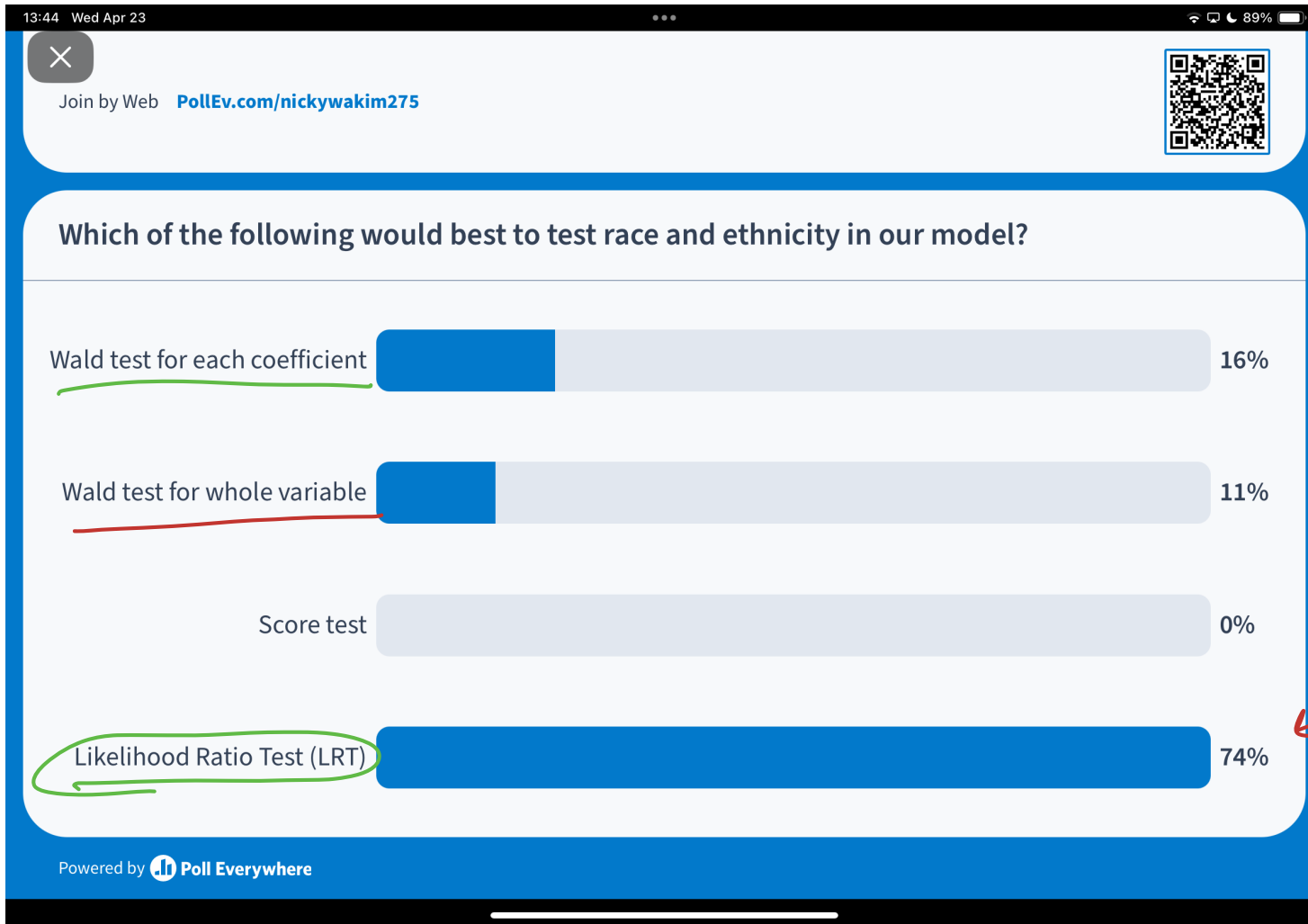
A note on wording

- When I say “test a single coefficient” or “test multiple coefficients” I am referring to the β 's
 - A single variable can have a single coefficient
 - Example: testing age
 - A single variable can have multiple coefficients
 - Example: testing race and ethnicity
 - Multiple variables will have multiple coefficients
 - Example: testing age and race and ethnicity together
- When I say “test a variable” I mean “determine if the model with the variable is more likely than the model without that variable”
 - We can use the Wald test to do this in some scenarios (i.e. single, continuous covariate)
 - BUT I advise you practice using the LRT whenever comparing models (aka testing variables)

All three tests together



Poll Everywhere Question 2



CORRECT

From Lesson 8: Wald test

- Assumes test statistic W follows a standard normal distribution under the null hypothesis
- Test statistic:

$$W = \frac{\hat{\beta}_j}{SE_{\hat{\beta}_j}} \sim N(0, 1)$$

standard normal

- where $\hat{\beta}_j$ is a MLE of coefficient j
- 95% Wald confidence interval:

$$\hat{\beta}_j \pm 1.96 \cdot SE_{\hat{\beta}_j}$$

- The Wald test is a routine output in R (`summary()` of `glm()` output)
 - Includes $SE_{\hat{\beta}_j}$ and can easily find confidence interval with `tidy()`
- Important note:** Wald test is best for confidence intervals of our coefficient estimates or estimated odds ratios.

Wald test in our example

- From our multiple logistic regression model:

$$\text{logit}(\pi(x_i)) = \beta_0 + \beta_1 \cdot I(R/E = \text{"H/L"}) + \beta_2 \cdot I(R/E = \text{"NH AIAN"}) \\ + \beta_3 \cdot I(R/E = \text{"NH API"}) + \beta_4 \cdot I(R/E = \text{"NH B"}) + \beta_5 \cdot \text{Age}_c$$

- Wald test can help us **construct the confidence interval for ALL coefficient estimates**
- If we want to use the **Wald test** to determine if a variable is significant in our model
 - Can only do so for **age**

Example 1: Single, continuous variable: Age

Single, continuous variable: Age

Given race and ethnicity is already in the model, is the regression model with age more likely than the model without age?

Needed steps:

1. Set the **level of significance** α
2. Specify the **null** (H_0) and **alternative** (H_A) hypotheses
3. Calculate the **confidence interval** and **determine if it overlaps with null**
4. Write a **conclusion** to the hypothesis test

Example 1: Single, continuous variable: Age

Single, continuous variable: Age

Given race and ethnicity is already in the model, is the regression model with age more likely than the model without age?

1. Set the **level of significance** α

- $\alpha = 0.05$

2. Specify the **null** (H_0) and **alternative** (H_A) hypotheses

- $H_0 : \beta_5 = 0$
- $H_1 : \beta_5 \neq 0$

Example 1: Single, continuous variable: Age

Single, continuous variable: Age

Given race and ethnicity is already in the model, is the regression model with age more likely than the model without age?

3. Calculate the **confidence interval** and **determine if it overlaps with null**

- Look at **coefficient estimates** (CI should not contain 0) OR **estimated odds ratio** (CI should not contain 1)

```
1 tidy(multi_bc, conf.int=T) %>%  
2   gt() %>%  
3   tab_options(table.font.size = 28) %>%  
4   fmt_number(decimals = 2)
```

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	-1.04	0.03	-38.05	0.00	-1.09	-0.99
Race_EthnicityHispanic-Latinx	-0.02	0.08	-0.18	0.85	-0.18	0.15
Race_EthnicityNH American Indian/Alaskan Native	-0.09	0.48	-0.18	0.86	-1.12	0.81
Race_EthnicityNH Asian/Pacific Islander	0.13	0.08	1.60	0.11	-0.03	0.30
Race_EthnicityNH Black	0.36	0.07	4.98	0.00	0.22	0.50
Age_c	0.06	0.00	17.81	0.00	0.05	0.06

$\exp(0) = 1$
 $e^0 = 1$

Example 1: Single, continuous variable: Age

Single, continuous variable: Age

Given race and ethnicity is already in the model, is the regression model with age more likely than the model without age?

3. Calculate the **confidence interval** and **determine if it overlaps with null**

- Look at *coefficient estimates* (CI should not contain 0) OR *estimated odds ratio* (CI should not contain 1)

```
1 tbl_regression(multi_bc,  
2               label = list(Race_Ethnicity ~ "Ra  
3                       Age_c ~ "Age (1 year  
4               exponentiate = TRUE) %>%  
5   as_gt() %>%  
6   tab_options(table.font.size = 30) %>%  
7   tab_source_note(md("*NH stands for \"Non-Hispa
```

Characteristic	OR ¹	95% CI ¹	p-value
Race/ethnicity			
NH White	—	—	
Hispanic-Latinx	0.98	0.83, 1.16	0.9
NH American Indian/Alaskan Native	0.92	0.33, 2.25	0.9
NH Asian/Pacific Islander	1.14	0.97, 1.35	0.11
NH Black	1.43	1.24, 1.65	<0.001
Age (1 year increase)	1.06	1.05, 1.07	<0.001

¹ OR = Odds Ratio, CI = Confidence Interval

NH stands for "Non-Hispanic"

Example 1: Single, continuous variable: Age

Single, continuous variable: Age

Given race and ethnicity is already in the model, is the regression model with age more likely than the model without age?

4. Write a **conclusion** to the hypothesis test

Given race and ethnicity is already in the model, the regression model with age is more likely than the model without age ($p\text{-val} < 0.001$).

Likelihood ratio test (1/3)

- Likelihood ratio test answers the question:
 - For a specific covariate, **which model tell us more about the outcome variable:** the model including the covariate (or set of covariates) or the model omitting the covariate (or set of covariates)?
 - Aka: Which model is more likely given our data: model including the covariate or the model omitting the covariate?
- Test a single coefficient by comparing different models
 - **Very similar to the F-test**
- Important: LRT can be used conduct hypothesis tests for **multiple coefficients**
 - Just like F-test, we can test a single coefficient, continuous/binary covariate, multi-level covariate, or multiple covariates

Likelihood ratio test (2/3)

- If testing **single variable** and it's **continuous or binary**, still use this hypothesis test:
 - $H_0: \beta_j = 0$
 - $H_1: \beta_j \neq 0$
- If testing **single variable** and it's **categorical with more than 2 groups**, use this hypothesis test:
 - $H_0: \beta_j = \beta_{j+1} = \dots = \beta_{j+i-1} = 0$
 - H_1 : at least one of the above β 's is not equal to 0
- If testing a **set of variables**, use this hypothesis test:
 - $H_0: \beta_1 = \beta_2 = \dots = \beta_k = 0$
 - H_1 : at least one of the above β 's is not equal to 0

Poll Everywhere Question 3

14:09 Wed Apr 23

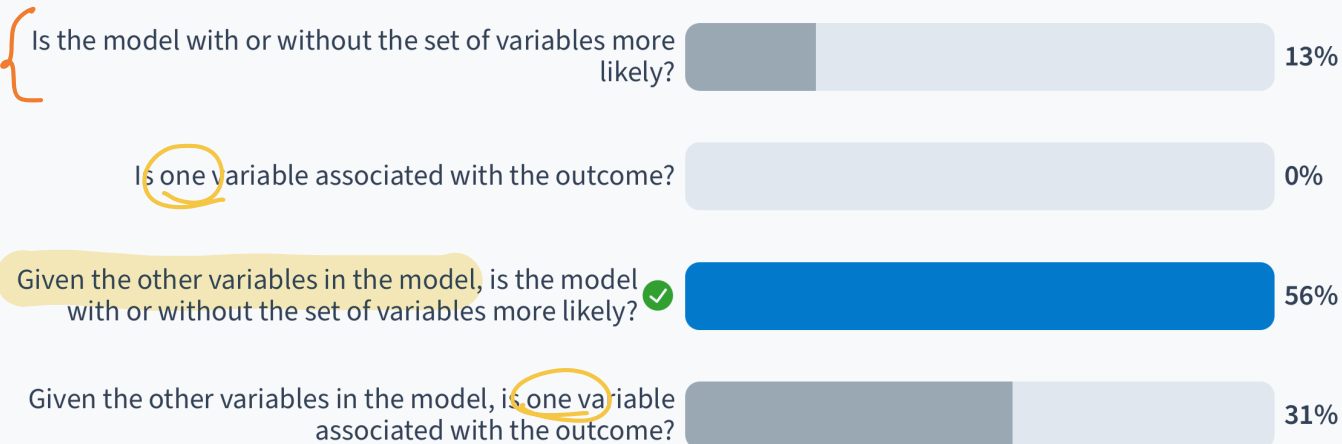
83%



Join by Web PollEv.com/nickywakim275



What's another way to think of our hypothesis test for a set of variables? How could we pose this in a written question? Hypothesis: $H_0 : \beta_j = \beta_{j+1} = \dots = \beta_{j+i-1} = 0$ vs H_1 : at least one of the above β 's is not equal to 0



Likelihood ratio test (3/3)

- From our multiple logistic regression model:

$$\text{logit}(\pi(\mathbf{X})) = \beta_0 + \beta_1 \cdot I(R/E = \text{"H/L"}) + \beta_2 \cdot I(R/E = \text{"NH AIAN"}) \\ + \beta_3 \cdot I(R/E = \text{"NH API"}) + \beta_4 \cdot I(R/E = \text{"NH B"}) + \beta_5 \cdot Age_c$$

- We can test a single coefficient or multiple coefficients
 - **Example 1:** Single, continuous variable: Age
 - **Example 2:** Single, >2 categorical variable: Race and Ethnicity
 - **Example 3:** Set of variables: Race and Ethnicity, and Age

Reminder on nested models



- Likelihood ratio test is only suitable to test “nested” models
- “Nested” models means the bigger model (full model) contains all the independent variables of the smaller model (reduced model)
- We cannot compare the following two models using LRT:

▪ Model 1:

$$\text{logit}(\pi(\mathbf{X})) = \beta_0 + \beta_1 \cdot I(R/E = \text{“H/L”}) + \beta_2 \cdot I(R/E = \text{“NH AIAN”}) \\ + \beta_3 \cdot I(R/E = \text{“NH API”}) + \beta_4 \cdot I(R/E = \text{“NH B”})$$

▪ Model 2:

Not nested

$$\text{logit}(\pi(Age)) = \beta_0 + \beta_1 \cdot Age_c$$

- If the two models to be compared are not nested, likelihood ratio test should not be used

Example 1: Single, continuous variable: Age

Single, continuous variable: Age

Given race and ethnicity is already in the model, is the regression model with age more likely than the model without age?

Needed steps:

1. Set the **level of significance** α
2. Specify the **null** (H_0) and **alternative** (H_A) hypotheses
3. Calculate the **test statistic** and **p-value**
4. Write a **conclusion** to the hypothesis test

Example 1: Single, continuous variable: Age

Single, continuous variable: Age

Given race and ethnicity is already in the model, is the regression model with age more likely than the model without age?

1. Set the **level of significance** α
 - $\alpha = 0.05$
2. Specify the **null** (H_0) and **alternative** (H_A) hypotheses
 - $H_0 : \beta_5 = 0$ or model without age is more likely
 - $H_1 : \beta_5 \neq 0$ or model with age is more likely

Example 1: Single, continuous variable: Age

Single, continuous variable: Age

Given race and ethnicity is already in the model, is the regression model with age more likely than the model without age?

3. Calculate the test statistic and p-value

```
1 multi_bc = glm(Late_stage_diag ~ Race_Ethnicity + Age_c, data = bc, family = binomial)
2 re_bc = glm(Late_stage_diag ~ Race_Ethnicity, data = bc, family = binomial)
3
4 lmtest::lrtest(multi_bc, re_bc)
```

Likelihood ratio test

Model 1: Late_stage_diag ~ Race_Ethnicity + Age_c

Model 2: Late_stage_diag ~ Race_Ethnicity

	#Df	LogLik	Df	Chisq	Pr(>Chisq)
1	6	-5741.8			
2	5	-5918.1	-1	352.63	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Example 1: Single, continuous variable: Age

Single, continuous variable: Age

Given race and ethnicity is already in the model, is the regression model with age more likely than the model without age?

4. Write a **conclusion** to the hypothesis test

Given race and ethnicity is already in the model, the regression model with age is more likely than the model without age ($p\text{-val} < 2.2 \cdot 10^{-16}$).

Poll Everywhere Question 4

14:16 Wed Apr 23

81%



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When testing a single continuous variable, like age, will the Wald test and LRT have the same results?

Always, yes.

6%

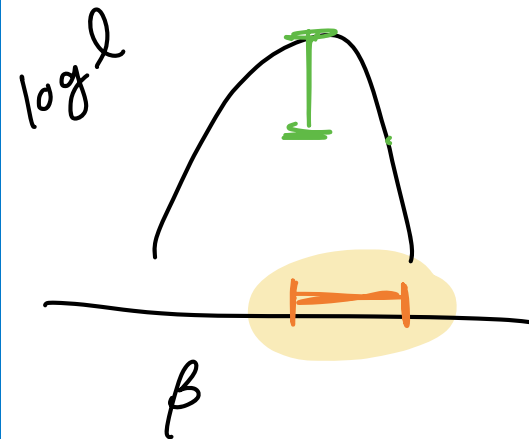
Most of the time, yes. The two tests are asymptotically equivalent. They only differ for small sample sizes. ✓

78%

Always, no.

17%

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Example 2: Single, >2 categorical variable: Race and Ethnicity

Single, >2 categorical variable: Race and Ethnicity

Given age is already in the model, is the regression model with race and ethnicity more likely than the model without race and ethnicity?

Needed steps:

1. Set the **level of significance** α
2. Specify the **null** (H_0) and **alternative** (H_A) **hypotheses**
3. Calculate the **test statistic** and **p-value**
4. Write a **conclusion** to the hypothesis test

Example 2: Single, >2 categorical variable: Race and Ethnicity

Single, >2 categorical variable: Race and Ethnicity

Given age is already in the model, is the regression model with race and ethnicity more likely than the model without race and ethnicity?

1. Set the **level of significance** α

- $\alpha = 0.05$

2. Specify the **null** (H_0) and **alternative** (H_A) hypotheses

- $H_0 : \beta_1 = \beta_2 = \beta_3 = \beta_4 = 0$ or model without race and ethnicity is more likely
- H_1 : at least one β is not 0 or model with race and ethnicity is more likely

Example 2: Single, >2 categorical variable: Race and Ethnicity

Single, >2 categorical variable: Race and Ethnicity

Given age is already in the model, is the regression model with race and ethnicity more likely than the model without race and ethnicity?

3. Calculate the test statistic and p-value

```
1 multi_bc = glm(Late_stage_diag ~ Race_Ethnicity + Age_c, data = bc, family = binomial)
2 age_bc = glm(Late_stage_diag ~ Age_c, data = bc, family = binomial)
3
4 lmtest::lrtest(multi_bc, age_bc)
```

Likelihood ratio test

Model 1: Late_stage_diag ~ Race_Ethnicity + Age_c

Model 2: Late_stage_diag ~ Age_c

	#Df	LogLik	Df	Chisq	Pr(>Chisq)
1	6	-5741.8			
2	2	-5754.8	-4	26.053	3.087e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Example 2: Single, >2 categorical variable: Race and Ethnicity

Single, >2 categorical variable: Race and Ethnicity

Given age is already in the model, is the regression model with race and ethnicity more likely than the model without race and ethnicity?

4. Write a **conclusion** to the hypothesis test

Given age is already in the model, the regression model with race and ethnicity is more likely than the model without race and ethnicity ($p\text{-val} = 3.1 \cdot 10^{-5} < 0.05$).

Example 3: Set of variables: Race and Ethnicity, and Age

Set of variables: Race and Ethnicity, and Age

Is the regression model with race and ethnicity and age more likely than the model without race and ethnicity nor age?

Needed steps:

1. Set the **level of significance** α
2. Specify the **null** (H_0) and **alternative** (H_A) hypotheses
3. Calculate the **test statistic** and **p-value**
4. Write a **conclusion** to the hypothesis test

Example 3: Set of variables: Race and Ethnicity, and Age

Set of variables: Race and Ethnicity, and Age

Is the regression model with race and ethnicity and age more likely than the model without race and ethnicity nor age?

1. Set the **level of significance** α

- $\alpha = 0.05$

2. Specify the **null** (H_0) and **alternative** (H_A) hypotheses

- $H_0 : \beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = 0$ or model without race and ethnicity and age is more likely
- H_1 : at least one β is not 0 or model with race and ethnicity and age is more likely

Example 3: Set of variables: Race and Ethnicity, and Age

Set of variables: Race and Ethnicity, and Age

Is the regression model with race and ethnicity and age more likely than the model without race and ethnicity nor age?

3. Calculate the test statistic and p-value

```
1 multi_bc = glm(Late_stage_diag ~ Race_Ethnicity + Age_c, data = bc, family = binomial)
2 intercept_bc = glm(Late_stage_diag ~ 1, data = bc, family = binomial)
3
4 lmtest::lrtest(multi_bc, intercept_bc)
```

Likelihood ratio test

Model 1: Late_stage_diag ~ Race_Ethnicity + Age_c

Model 2: Late_stage_diag ~ 1

	#Df	LogLik	Df	Chisq	Pr(>Chisq)
1	6	-5741.8			
2	1	-5930.5	-5	377.32	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Example 3: Set of variables: Race and Ethnicity, and Age

Set of variables: Race and Ethnicity, and Age

Is the regression model with race and ethnicity and age more likely than the model without race and ethnicity nor age?

4. Write a **conclusion** to the hypothesis test

The regression model with race and ethnicity and age is more likely than the model omitting race and ethnicity and age ($p\text{-val} < 2.2 \cdot 10^{-16}$).

Learning Objectives

1. Construct and fit a multiple logistic regression model
2. Test for significance of individual coefficients or sets of coefficients in multiple logistic regression
3. Estimate the predicted probability of our outcome using multiple logistic regression
4. Present the odds ratios for multiple variables at once
5. Interpret odds ratios for coefficients while adjusting for other variables

Estimated/Predicted Probability for MLR

- Basic idea for predicting/estimating probability stays the same
- Calculations will be slightly different
 - Especially for the confidence interval
- Recall our fitted model for late stage breast cancer diagnosis:

$$\text{logit}(\hat{\pi}(\mathbf{X})) = -4.56 - 0.02 \cdot I(R/E = \text{"H/L"}) - 0.09 \cdot I(R/E = \text{"NH AIAN"}) \\ + 0.13 \cdot I(R/E = \text{"NH API"}) + 0.36 \cdot I(R/E = \text{"NH B"}) + 0.06 \cdot Age_c$$

Predicted Probability

- We may be interested in predicting probability of having a late stage breast cancer diagnosis for a specific age.
- The predicted probability is the estimated probability of having the event for given values of covariate(s)
- Recall our fitted model for late stage breast cancer diagnosis:

$$\text{logit}(\hat{\pi}(\mathbf{X})) = -4.56 + 0.02 \cdot I(R/E = \text{"H/L"}) - 0.09 \cdot I(R/E = \text{"NH AIAN"}) + 0.13 \cdot I(R/E = \text{"NH API"}) + 0.36 \cdot I(R/E = \text{"NH B"}) + 0.06 \cdot \text{Age}_c$$

- We can convert it to the predicted probability:

$$\rightarrow \hat{\pi}(\mathbf{X}) = \frac{\exp(\hat{\beta}_0 + \hat{\beta}_1 \cdot I(R/E = \text{"H/L"}) + \dots + \hat{\beta}_5 \cdot \text{Age}_c)}{1 + \exp(\hat{\beta}_0 + \hat{\beta}_1 \cdot I(R/E = \text{"H/L"}) + \dots + \hat{\beta}_5 \cdot \text{Age}_c)}$$

$\rightarrow \frac{\exp(\dots)}{\exp(\dots)} = e$

- This is an inverse logit calculation
- We can calculate this using the `predict()` function like in BSTA 512
 - Another option: taking inverse logit of fitted values from `augment()` function

Predicting probability in R

Predicting probability of late stage breast cancer diagnosis

For someone who is 60 years old and Non-Hispanic Asian/Pacific Islander, what is the predicted probability for late stage breast cancer diagnosis (with confidence intervals)?

Needed steps:

1. Calculate probability prediction
2. Check if we can use Normal approximation
3. Calculate confidence interval
 - a. Using logit scale then converting
 - b. Using Normal approximation
4. Interpret results

Predicting probability in R

Predicting probability of late stage breast cancer diagnosis

For someone who is 60 years old and Non-Hispanic Asian/Pacific Islander, what is the predicted probability for late stage breast cancer diagnosis (with confidence intervals)?

1. Calculate probability prediction

```
1 newdata = data.frame(Age_c = 60 - mean_age,  
2                       Race_Ethnicity = "NH Asian/Pacific Islander")  
3 pred1 = predict(multi_bc, newdata, se.fit = T, type="response")  
4 pred1
```

\$fit

1
0.2685667

\$se.fit

1
0.01572695

\$residual.scale

[1] 1

↳ fitted from glm()
↳ probability
"link" = logit

Predicting probability in R

Predicting probability of late stage breast cancer diagnosis

For someone who is 60 years old and Non-Hispanic Asian/Pacific Islander, what is the predicted probability for late stage breast cancer diagnosis (with confidence intervals)?

2. Check if we can use Normal approximation

We can use the Normal approximation if: $\hat{p}n = \hat{\pi}(X) \cdot n > 10$ and $(1 - \hat{p})n = (1 - \hat{\pi}(X)) \cdot n > 10$.

```
1 n = nobs(multi_bc)
2 p = pred1$fit
3 n*p
```

2685.667 > 10

```
1 n*(1-p)
```

7314.333 > 10

We can use the Normal approximation!

Predicting probability in R

Predicting probability of late stage breast cancer diagnosis

For someone who is 60 years old and Non-Hispanic Asian/Pacific Islander, what is the predicted probability for late stage breast cancer diagnosis (with confidence intervals)?

3b. Calculate confidence interval (Option 2: with Normal approximation)

```
1 pred = predict(multi_bc, newdata, se.fit = T, type = "response")
2
3 LL_CI = pred$fit - qnorm(1-0.05/2) * pred$se.fit
4 UL_CI = pred$fit + qnorm(1-0.05/2) * pred$se.fit
5
6 c(Pred = pred$fit, LL_CI, UL_CI) %>% round(digits=3)
```

Pred.1	1	1
0.269	0.238	0.299

see lesson 7
more detail
on this

Predicting probability in R

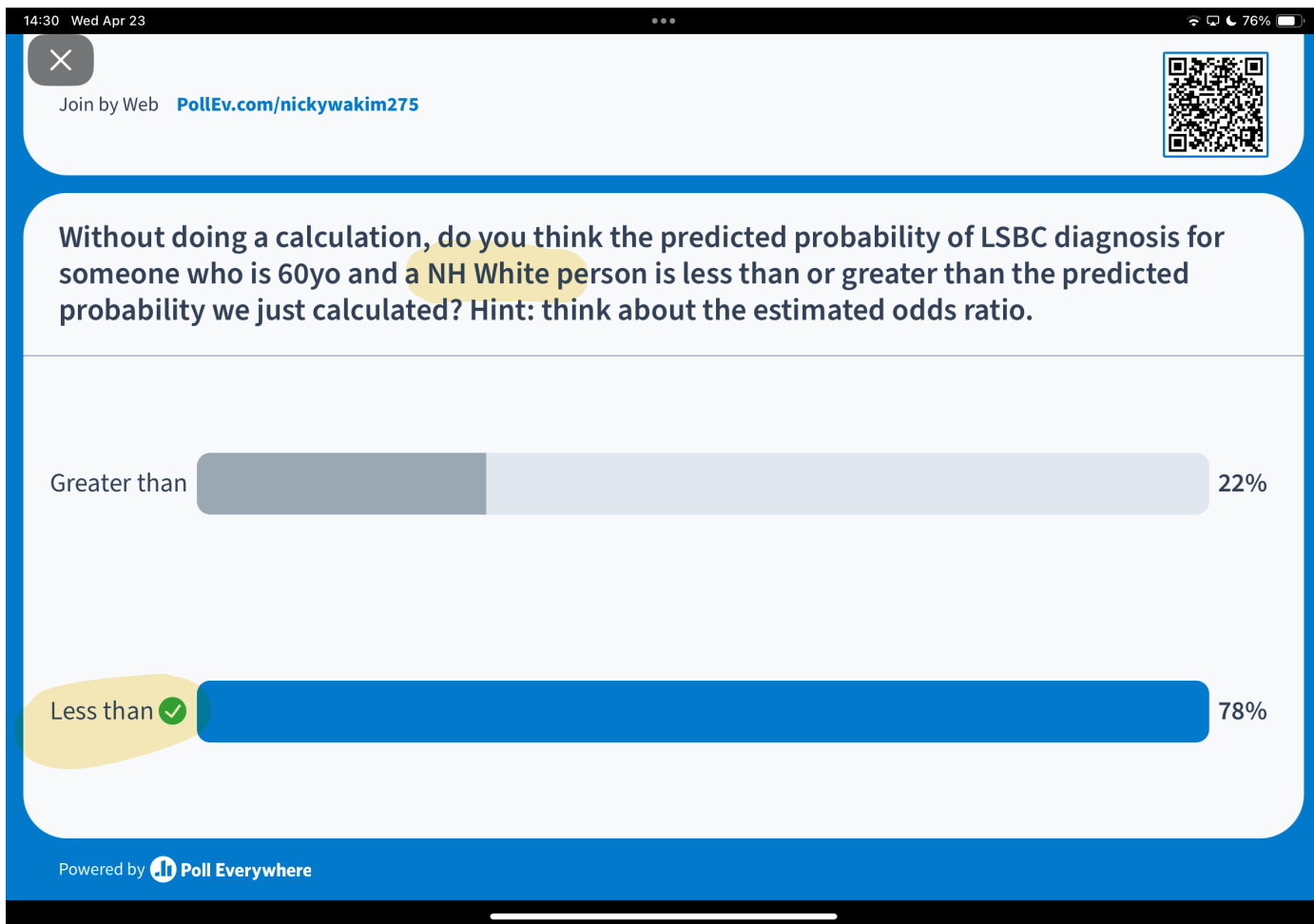
Predicting probability of late stage breast cancer diagnosis

For someone who is 60 years old and Non-Hispanic Asian/Pacific Islander, what is the predicted probability for late stage breast cancer diagnosis (with confidence intervals)?

4. Interpret results

For someone who is 60 years old and Non-Hispanic Asian/Pacific Islander, the predicted probability of late stage breast cancer diagnosis is 0.269 (95% CI: 0.238, 0.299).

Poll Everywhere Question 5



$$\hat{OR} = 1.14$$

comparing
NH A / P1
to NH White

$odds_{NHAP1} \Rightarrow$

$odds_{NHW}$

$$\Rightarrow \hat{\pi}(NHAP1) > \hat{\pi}(NHW)$$

Learning Objectives

1. Construct and fit a multiple logistic regression model
2. Test for significance of individual coefficients or sets of coefficients in multiple logistic regression
3. Estimate the predicted probability of our outcome using multiple logistic regression
4. Present the odds ratios for multiple variables at once
5. Interpret odds ratios for coefficients while adjusting for other variables

How to present odds ratios: Table

```
1 library(gtsummary)
2 tbl_regression(multi_bc,
3               label = list(Race_Ethnicity ~ "Race/ethnicity",
4                           Age_c ~ "Age (1 year increase)"),
5               exponentiate = TRUE) %>%
6 as_gt() %>%
7 tab_options(table.font.size = 38) %>%
8 tab_source_note(md("*NH stands for \"Non-Hispanic\"*"))
```

Characteristic	OR ¹	95% CI ¹	p-value
Race/ethnicity			
NH White	—	—	
Hispanic-Latinx	0.98	0.83, 1.16	0.9
NH American Indian/Alaskan Native	0.92	0.33, 2.25	0.9
NH Asian/Pacific Islander	1.14	0.97, 1.35	0.11
NH Black	1.43	1.24, 1.65	<0.001
Age (1 year increase)	1.06	1.05, 1.07	<0.001

¹ OR = Odds Ratio, CI = Confidence Interval

NH stands for "Non-Hispanic"

How to present odds ratios: Forest Plot Setup

```
1 library(broom.helpers)
2 MLR_tidy = tidy_and_attach(multi_bc, conf.int=T, exponentiate = T) %>%
3   tidy_remove_intercept() %>%
4   tidy_add_reference_rows() %>%
5   tidy_add_estimate_to_reference_rows() %>%
6   tidy_add_term_labels()
7 glimpse(MLR_tidy)
```

Rows: 6

Columns: 16

\$ term	<chr> "Race_EthnicityNH White", "Race_EthnicityHispanic-Latin...
\$ variable	<chr> "Race_Ethnicity", "Race_Ethnicity", "Race_Ethnicity", "...
\$ var_label	<chr> "Race_Ethnicity", "Race_Ethnicity", "Race_Ethnicity", "...
\$ var_class	<chr> "factor", "factor", "factor", "factor", "factor", "nume...
\$ var_type	<chr> "categorical", "categorical", "categorical", "categoric...
\$ var_nlevels	<int> 5, 5, 5, 5, 5, NA
\$ contrasts	<chr> "contr.treatment", "contr.treatment", "contr.treatment"...
\$ contrasts_type	<chr> "treatment", "treatment", "treatment", "treatment", "tr...
\$ reference_row	<lgl> TRUE, FALSE, FALSE, FALSE, FALSE, NA
\$ label	<chr> "NH White", "Hispanic-Latinx", "NH American Indian/Alas...
\$ estimate	<dbl> 1.0000000, 0.9846940, 0.9178662, 1.1433526, 1.4300256, ...
\$ std.error	<dbl> NA, 0.083653090, 0.484110085, 0.083796726, 0.071788616, ...
\$ statistic	<dbl> NA, -0.1843845, -0.1770333, 1.5986877, 4.9825778, 17.81...
\$ p.value	<dbl> NA, 8.537118e-01, 8.594822e-01, 1.098900e-01, 6.274274e...

How to present odds ratios: Forest Plot Setup

```
1 MLR_tidy = MLR_tidy %>%  
2   mutate(var_label = case_match(var_label,  
3     "Race_Ethnicity" ~ "Race and ethnicity",  
4     "Age_c" ~ ""),  
5     label = case_match(label,  
6       "NH White" ~ "Non-Hispanic White",  
7       "Hispanic-Latinx" ~ "Hispanic-Latinx",  
8       "NH American Indian/Alaskan Native" ~ "Non-Hispanic American \n Indian/Alaskan Native",  
9       "NH Asian/Pacific Islander" ~ "Non-Hispanic \n Asian/Pacific Islander",  
10      "NH Black" ~ "Non-Hispanic Black",  
11      "Age_c" ~ "Age (yrs)"))
```

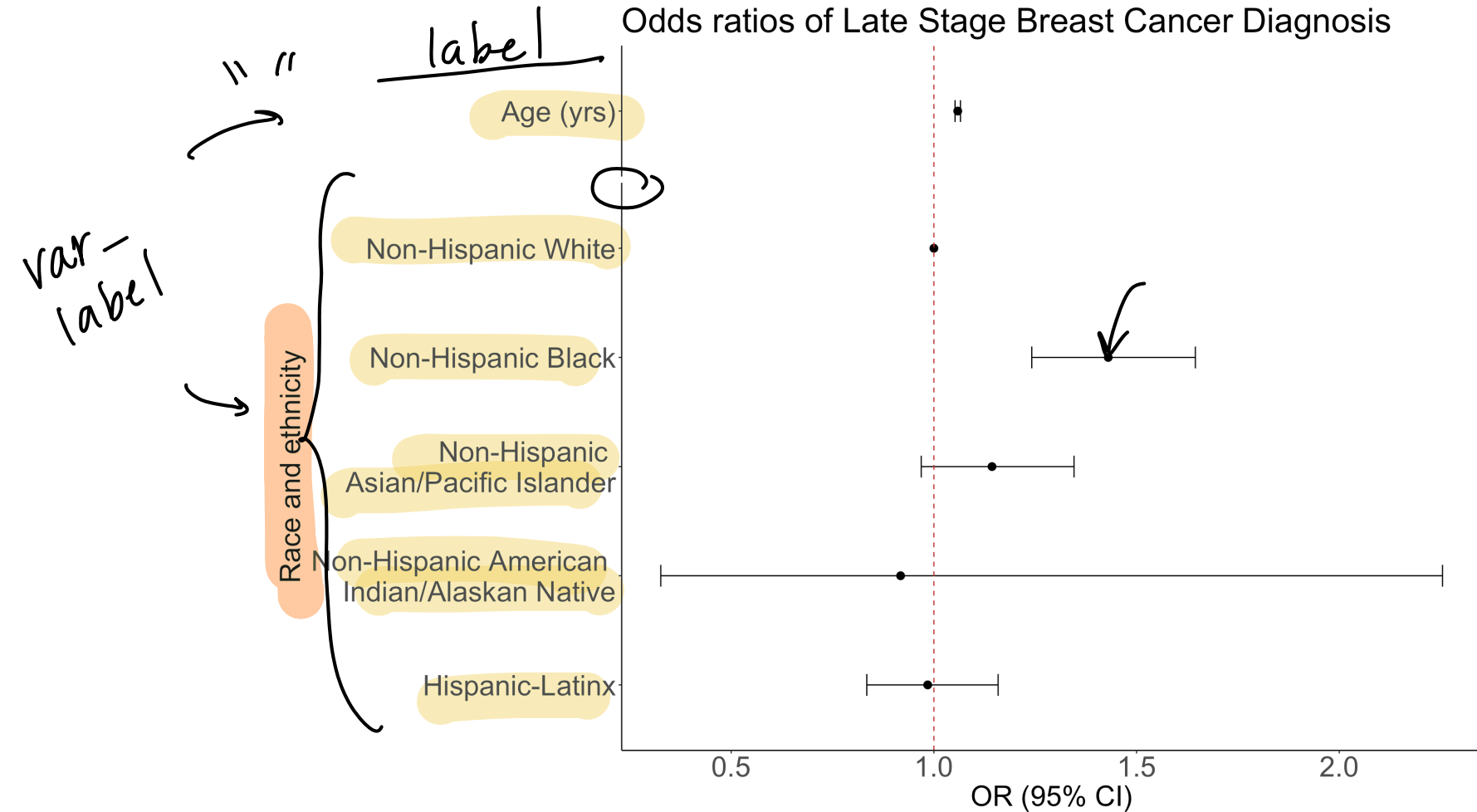
How to present odds ratios: Forest Plot

```
1 MLR_tidy = MLR_tidy %>% mutate(label = fct_reorder(label, term))
2
3 plot_MLR = ggplot(data=MLR_tidy,
4   aes(y=label, x=estimate, xmin=conf.low, xmax=conf.high)) +
5   geom_point(size = 3) + geom_errorbarh(height=.2) +
6
7   geom_vline(xintercept=1, color='#C2352F', linetype='dashed', alpha=1) +
8   theme_classic() +
9
10  facet_grid(rows = vars(var_label), scales = "free",
11             space='free_y', switch = "y") +
12
13  {
14    labs(x = "OR (95% CI)",
15         title = "Odds ratios of Late Stage Breast Cancer Diagnosis") +
16    theme(axis.title = element_text(size = 25),
17          axis.text = element_text(size = 25),
18          title = element_text(size = 25),
19          axis.title.y=element_blank(),
20          strip.text = element_text(size = 25),
21          strip.placement = "outside",
22          strip.background = element_blank())
23  }
```

groups by var_label → red dashed line

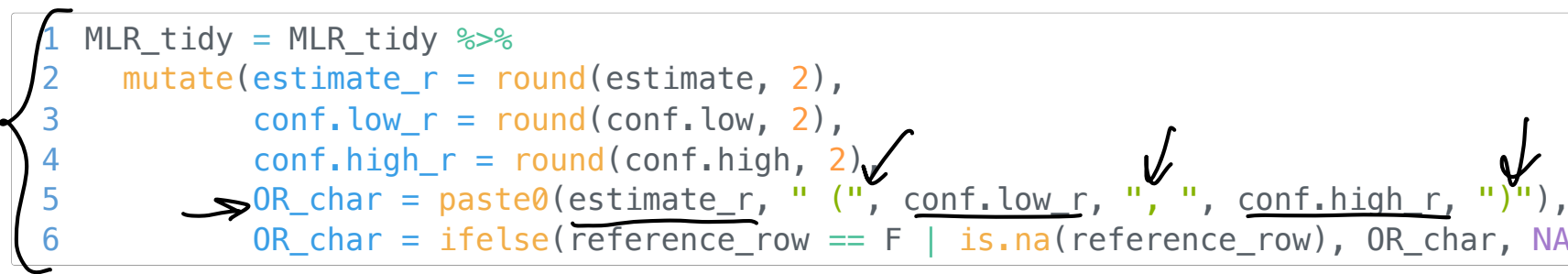
axes labels
size

How to present odds ratios: Forest Plot



Adding written estimates for odds ratios

```
1 MLR_tidy = MLR_tidy %>%  
2   mutate(estimate_r = round(estimate, 2),  
3         conf.low_r = round(conf.low, 2),  
4         conf.high_r = round(conf.high, 2),  
5         OR_char = paste0(estimate_r, " (", conf.low_r, ", ", conf.high_r, ")"),  
6         OR_char = ifelse(reference_row == F | is.na(reference_row), OR_char, NA))
```

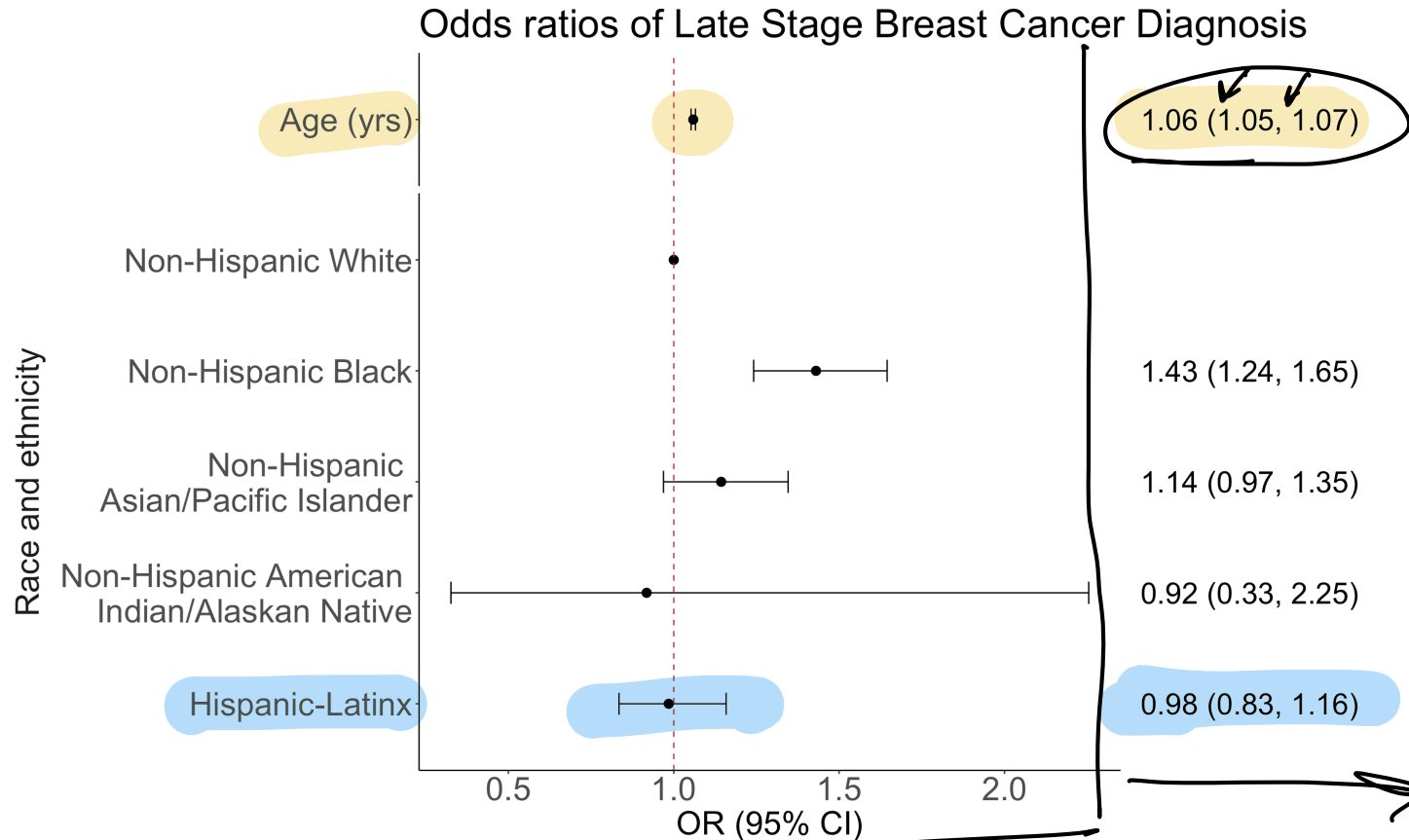


- “Plot” of the text for odds ratios estimates

```
1 OR_labs = ggplot(data=MLR_tidy, aes(y=label)) +  
2   geom_text(aes(x = -1, label = OR_char), hjust = 0, size=8) +  
3   xlim(-1, 1) +  
4   facet_grid(rows = vars(var_label), scales = "free",  
5             space='free_y') +  
6   theme_void() +  
7   theme(strip.text = element_blank())
```

Combine them!!

```
1 library(cowplot)
2 plot_grid(plot_MLR, OR_labs, ncol=2, align = "h", rel_widths = c(4, 1))
```



Learning Objectives

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Multivariable Logistic Regression Model

- The multivariable model of logistic regression (called multiple logistic regression) is useful in that it statistically adjusts the estimated effect of each variable in the model
- Each estimated coefficient provides an estimate of the log odds *adjusting for all other variables included in the model*
- The adjusted odds ratio can be different from or similar to the unadjusted odds ratio
- Comparing adjusted vs. unadjusted odds ratios can be a useful activity

↳ simple log. reg.

Interpretation of Coefficients in MLR

- The interpretation of coefficients in multiple logistic regression is *essentially the same as the interpretation of coefficients in simple logistic regression*
- For interpretation, we need to
 - point out that these are *adjusted estimates*
 - provide *a list of other variables in the model*

Example: Race and Ethnicity and Age model fit

Characteristic	OR ¹	95% CI ¹	p-value
Race/ethnicity			
NH White	—	—	
Hispanic-Latinx	0.98	0.83, 1.16	0.9
NH American Indian/Alaskan Native	0.92	0.33, 2.25	0.9
NH Asian/Pacific Islander	1.14	0.97, 1.35	0.11
NH Black	1.43	1.24, 1.65	<0.001
Age (1 year increase)	1.06	1.05, 1.07	<0.001

¹ OR = Odds Ratio, CI = Confidence Interval

- The estimated odds of late stage breast cancer diagnosis for Hispanic-Latinx individuals is 0.98 times that of Non-Hispanic White individuals, controlling for age (95% CI: 0.83, 1.16).
- The estimated odds of late stage breast cancer diagnosis for Non-Hispanic American Indian/Alaskan Natives is 0.92 times that of Non-Hispanic White individuals, controlling for age (95% CI: 0.33, 2.25).
- The estimated odds of late stage breast cancer diagnosis for Non-Hispanic Asian/Pacific Islanders is 1.14 times that of Non-Hispanic White individuals, controlling for age (95% CI: 0.97, 1.35).
- The estimated odds of late stage breast cancer diagnosis for Non-Hispanic Black individuals is 1.43 times that of Non-Hispanic White individuals, controlling for age (95% CI: 1.24, 1.65).
- For every one year increase in age, there is an 6% increase in the estimated odds of late stage breast cancer diagnosis, adjusting for race and ethnicity (95% CI: 5%, 7%).

can say
"controlling for"
OR
"adjusting for" →